

Open Source Information (OSINT) Petrochemical Public Tracking w/ AI Integration

Why do you want to pursue this idea?

Petrochemical companies are notorious for opaque corporate structures, aggressive subsidiary activity, and city-level regulatory lobbying and activity. Most tracking options focus on finance or industry grade data, accompanied by massive price tags. However, open source data on local laws, corporate filings, etc., is widely available through free APIs, public databases, and corporate websites.

Market: What exists out there like your idea?

There are virtually zero free, open-source tools that exist like this for everyday consumer, meaning climate damage, energy exports, and regulatory shifts happen invisible to consumer's eyes.

Innovation: What is the innovation in your idea? What is different about it?

Most existing supply chain and environmental monitoring platforms (like Kpler or S&P Global) are enterprise products locked behind paywalls. The primary innovation of this project is an open-source, multi-modal OSINT aggregation that correlates non-standardized public data streams into information completely free of charge.

- Multi-Modal Data Fusion (Cross-Domain Correlation)
- Automated Entity Resolution Detection (Theoretically lol)

Pytorch and open-source LLMs are a perfect fit for this project, alongside platforms like Streamlit. These are also all concepts that are extremely important to researchers in this field.

This project is very technically viable, and almost 100% free! Especially if the scope of the project stay relatively local (East Lansing, or maybe a Petrochemical hub in the US?) It also provides critical training in key concepts of data science, AI, and Machine Learning.

SEC EDGAR API

OpenCorporates API

DLA PUB LOG

EPA ECHO API (Enforcement and Compliance History Online)

OpenSky Network API

AIS Vessel Tracking (petrochemical shipping)

Sentinel Hub / Copernicus Open Access Hub

Team

The team for this project can be everyone from beginners to advanced AI/ML engineers, though at least a few team members should understand PyTorch and API connection. Though in my opinion it is beginner friendly, as StreamLit and data management are fun and relatively simple to learn!